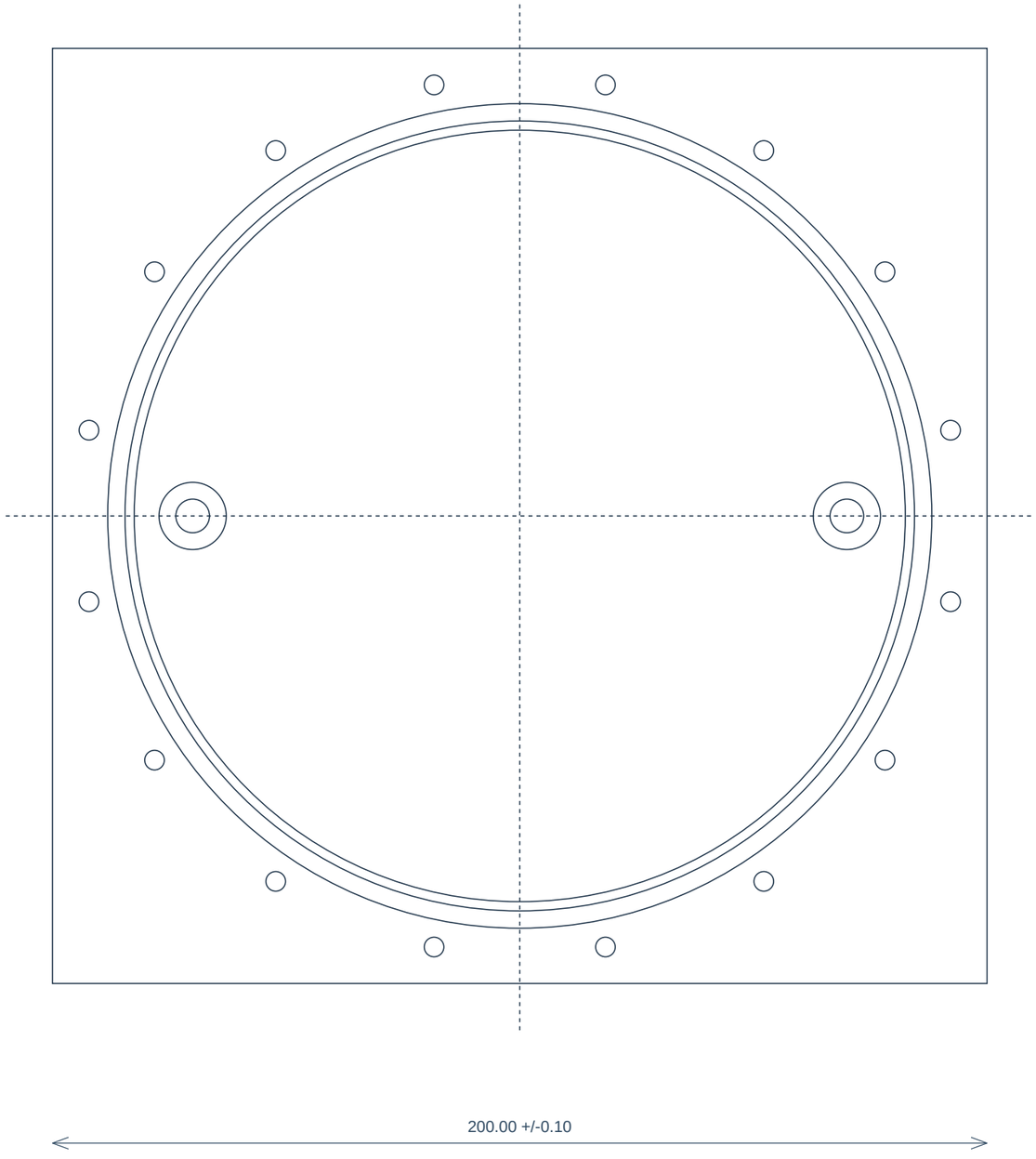
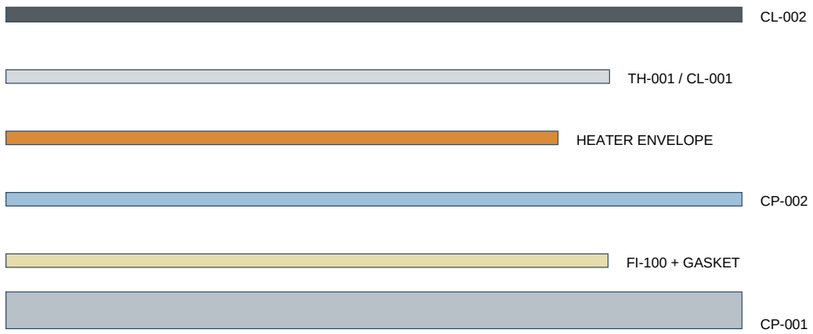


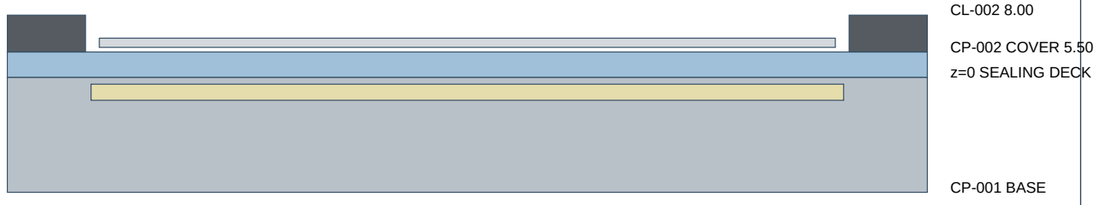
NOTES

1. CP-001 upper sealing deck z=0; cavity center x=y=0; ports on x-axis.
2. Assembly shown with FI-100. FI-000, FI-200 and FI-300 exchange without changing the base, cover or transfer interface.
3. Thirty kPa gauge is an analysis load case only, not a certified rating.
4. No heater or sensor holes penetrate CP-002. Heater envelope, coating, sensors and interface resistance remain unresolved.
5. Camera line of sight to the centered 150 diameter ROI is unobstructed.

EXPLODED VIEW (SCHEMATIC)



SECTION A-A (ASSEMBLED)

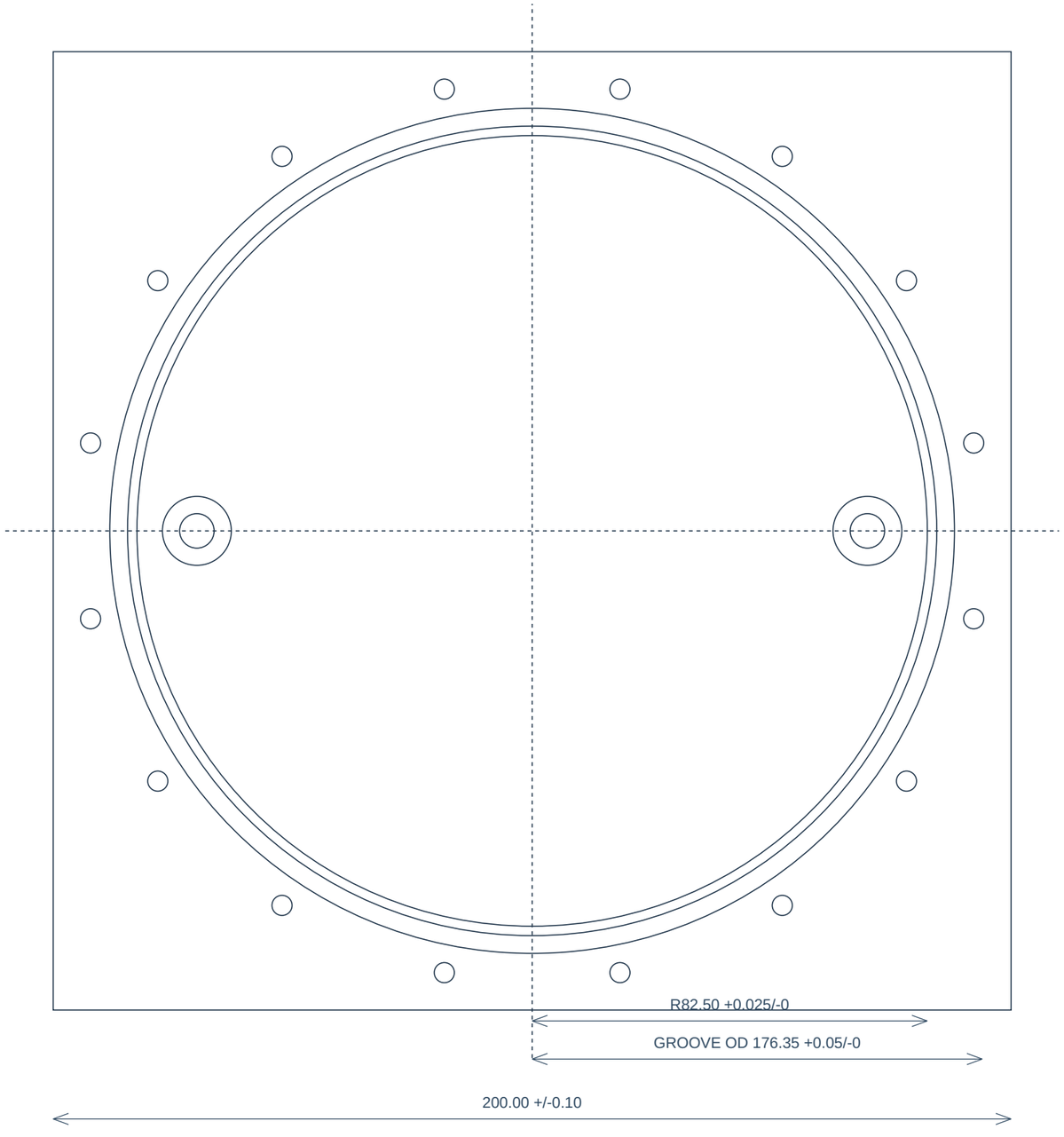


CP-REV-A GENERAL ARRANGEMENT

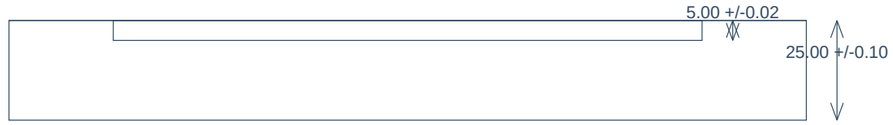
STATUS: PRELIMINARY ENGINEERING - NOT MACHINING APPROVED

SHEET 1/6 SCALE NTS UNITS mm

REV A INSPECTION NEAR 20 C



SECTION A-A



SIDE PORT AXES y=0, z=-14; candidate 1/4-18 NPT

INTERNAL DRILLS 6 nominal; vertical transfers x=+/-70

TRANSFER GLANDS: OD 14.38, width 2.50, depth 1.35

NOTES

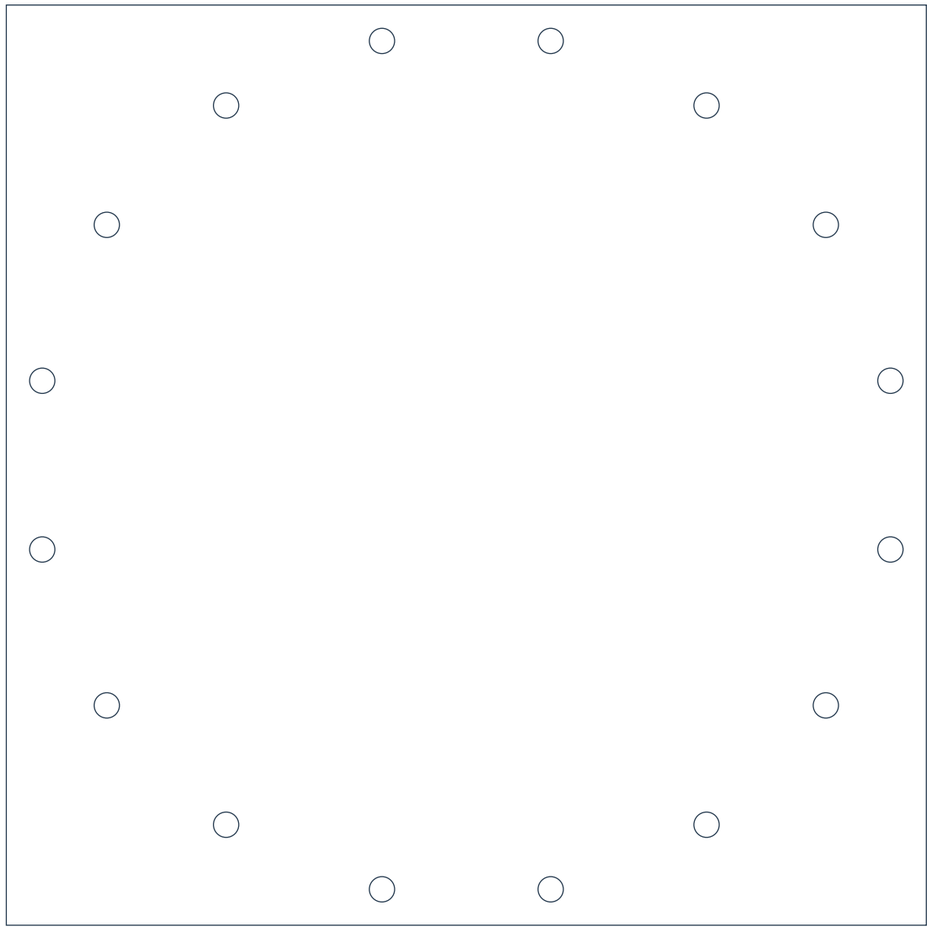
1. Material 6061 aluminum; use stress-relieved stock where available. Starting stock 203.2 x 203.2 x 25.4 nominal.
2. Deck flatness target 0.02; pocket-floor flatness target 0.03. Inspect without artificially flattening part.
3. Main gland width 3.70 +/-0.05, depth 2.12 +/-0.02, bottom R0.30 nominal, axis position diameter 0.05 relative to cavity axis.
4. Sealing face Ra <=0.8 micrometre; gland Ra <=1.6 micrometre.
5. Two static taps are modeled from the front edge to the defined transfer centers. Select actual fittings before release.

CP-001 BASE - DIMENSIONED PRELIMINARY

STATUS: PRELIMINARY ENGINEERING - NOT MACHINING APPROVED

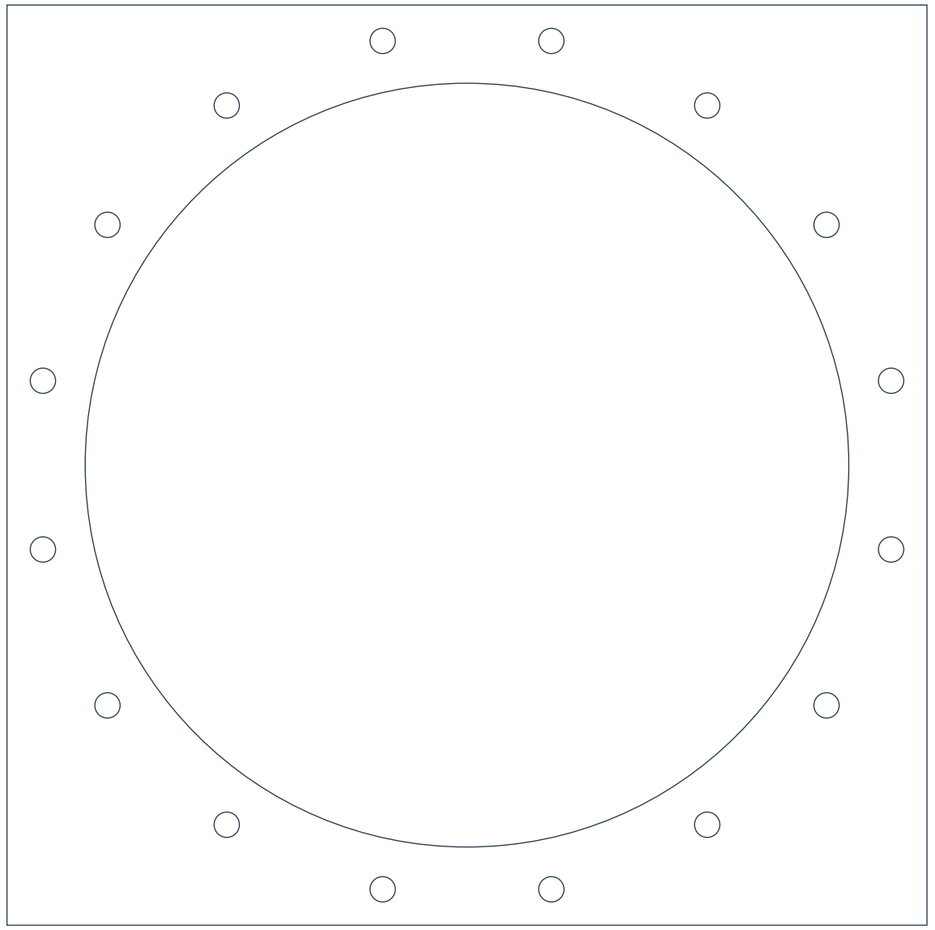
RELEASE ITEMS OPEN - SEE VALIDATION REPORT

CP-002 COVER



PCD 188 BASIC - 16 EQ SP @ 22.5 deg; OFFSET 11.25 deg

CL-002 STRUCTURAL CLAMP



NOTES

1. Cover 200 x 200 x 5.50 +/-0.05; water-facing flatness 0.03; face parallelism 0.03.
2. Clamp 200 x 200 x 8.00 +/-0.10; aperture diameter 166.00 +0.10/-0.00.
3. Both parts: clearance holes diameter 5.50 +0.10/-0.00; position diameter 0.10 to assembly datums.
4. Modeled M5x25 plus 1 mm washer stack gives 10.25-10.75 engagement. Verify actual screw, washer, thread runout and drill point.
5. Tightening torque is intentionally unspecified pending material, lubrication, preload and gasket-reaction definition.

CP-002 COVER AND CL-002 STRUCTURAL CLAMP

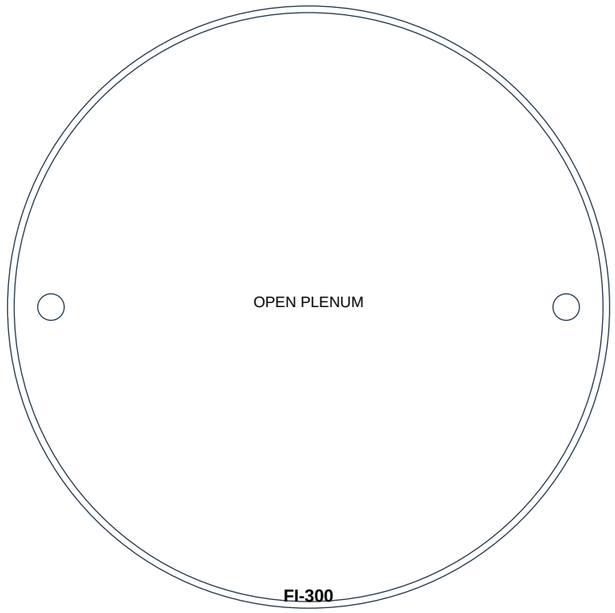
STATUS: PRELIMINARY ENGINEERING - NOT MACHINING APPROVED

SHEET 3/6 SCALE NTS UNITS mm

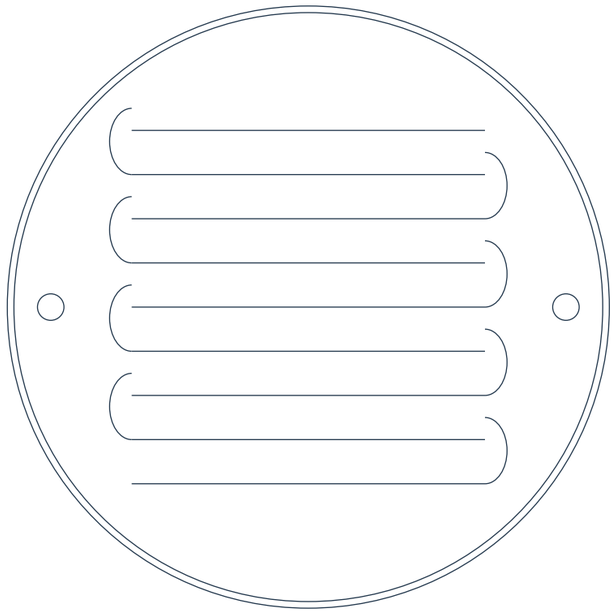
REV A INSPECTION NEAR 20 C

RELEASE ITEMS OPEN - SEE VALIDATION REPORT

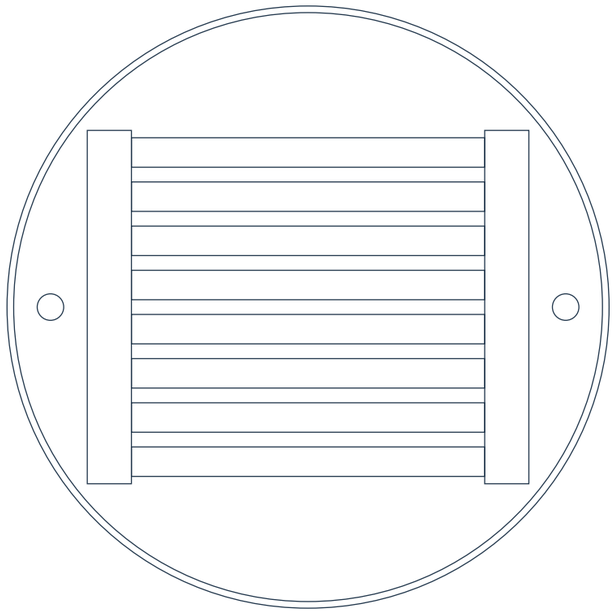
FI-000



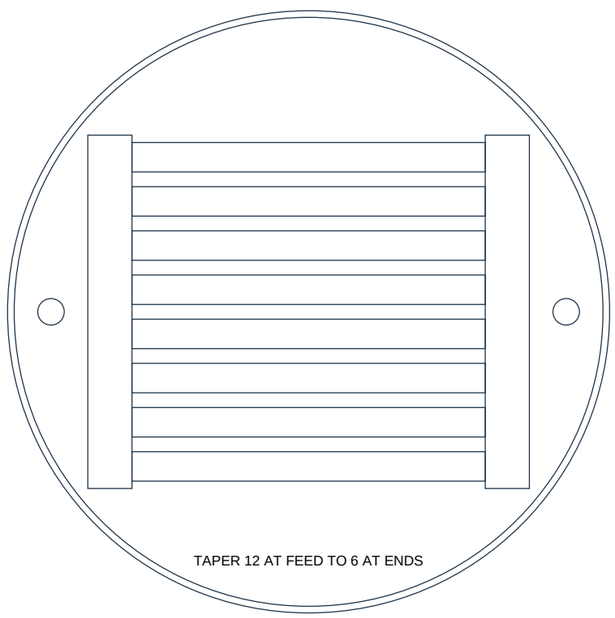
FI-100



FI-200



FI-300



NOTES

1. Common circumscribed insert OD 163.60 +/-0.05; nominal floor 1.00 +/-0.03; machined POM-C final candidate.
2. PETG meshes are prototypes and do not inherit machined-part tolerances.
3. Round locator at y=+70 and radial slot at y=-70 control rotation while permitting radial expansion about center.
4. FI-100 nominal CAD centerline length is reported in the validation report; it is not assumed to be 1.5 m.
5. FI-300 remains a tapered-manifold candidate until hydraulic modeling demonstrates improvement.
6. All variants use isolated transfer face seals, a perimeter rib seal and positively registered internal strip seals.

COMMON INSERT INTERFACE AND FLOW VARIANTS

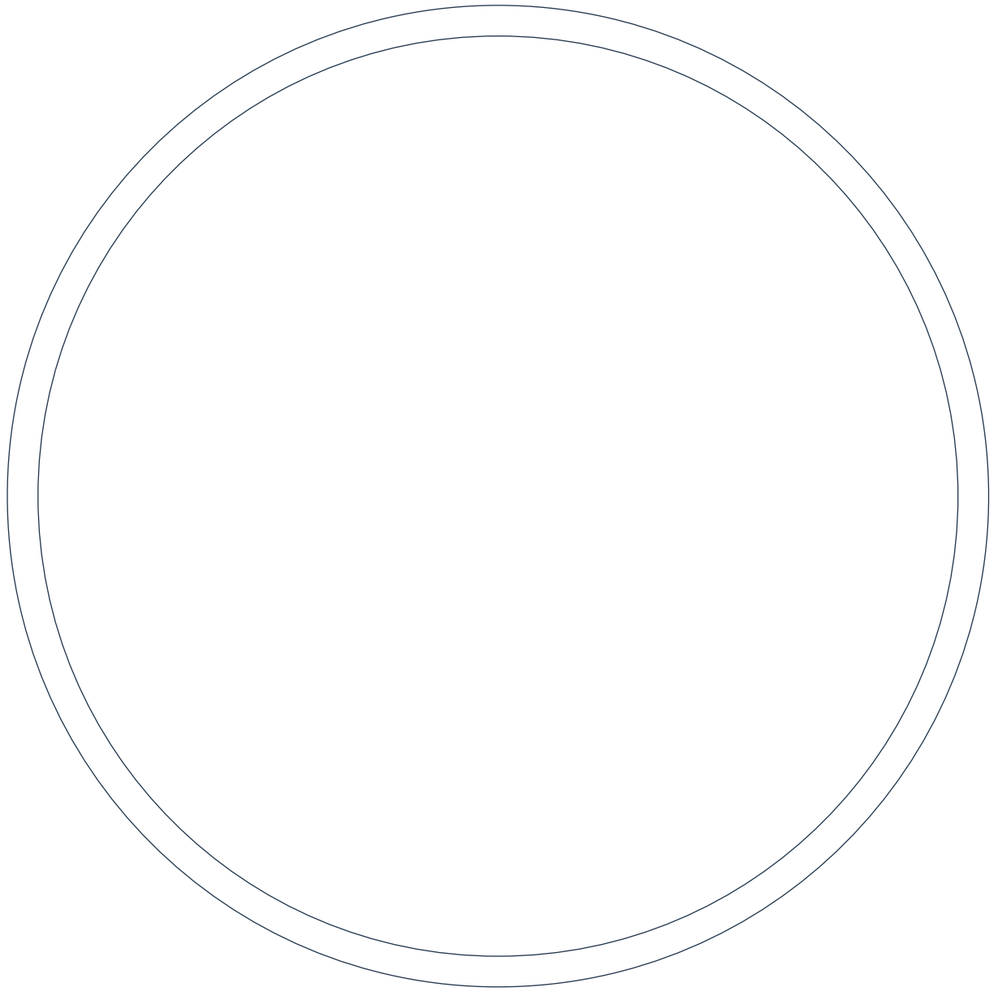
STATUS: PRELIMINARY ENGINEERING - NOT MACHINING APPROVED

SHEET 4/6 SCALE NTS UNITS mm

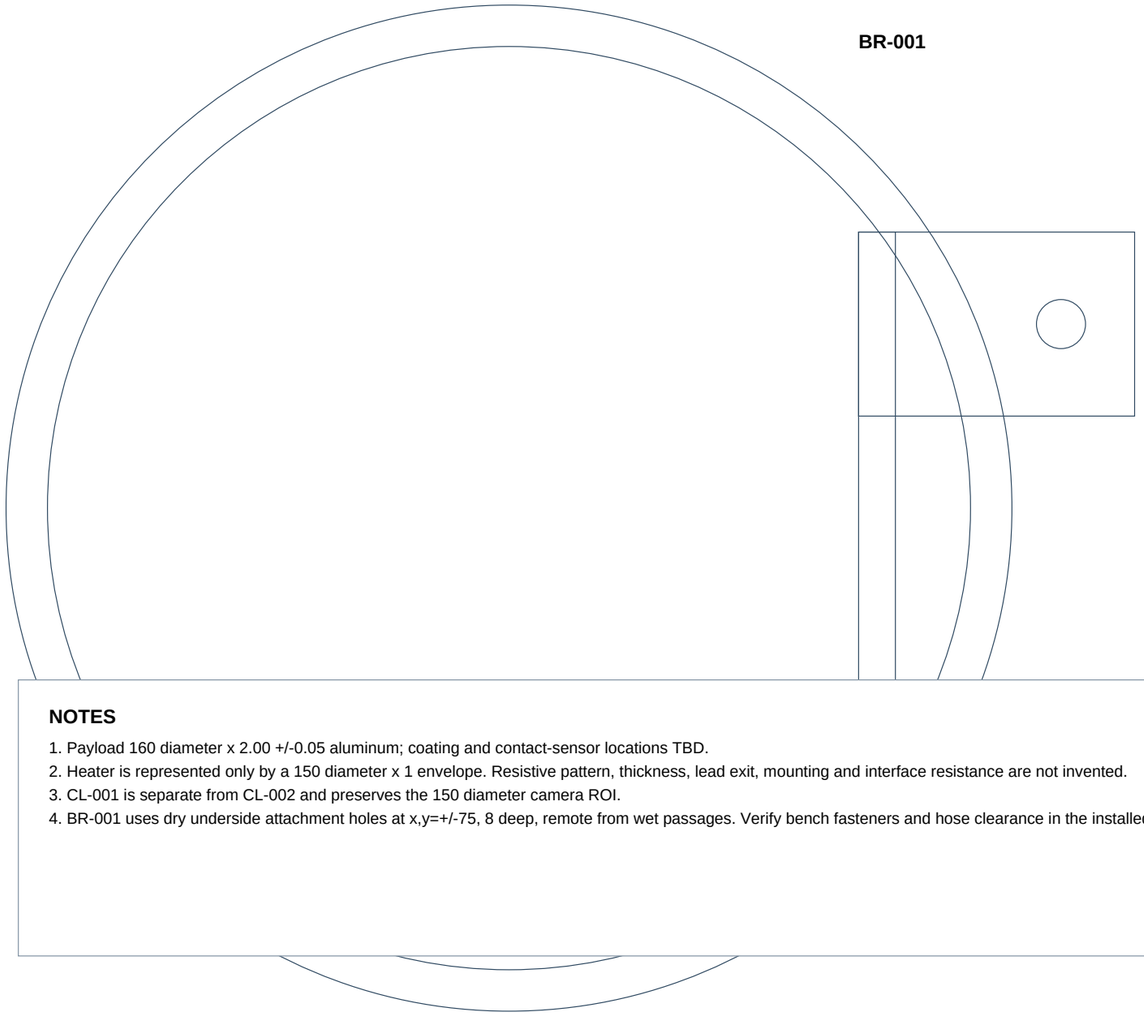
REV A INSPECTION NEAR 20 C

RELEASE ITEMS OPEN - SEE VALIDATION REPORT

TH-001 PAYLOAD / ROI



CL-001 PROVISIONAL CLAMP



OD 164; ID 150.5; thickness 4; 3 adjustment slots

NOTES

1. Payload 160 diameter x 2.00 +/-0.05 aluminum; coating and contact-sensor locations TBD.
2. Heater is represented only by a 150 diameter x 1 envelope. Resistive pattern, thickness, lead exit, mounting and interface resistance are not invented.
3. CL-001 is separate from CL-002 and preserves the 150 diameter camera ROI.
4. BR-001 uses dry underside attachment holes at x,y=+/-75, 8 deep, remote from wet passages. Verify bench fasteners and hose clearance in the installed rig.

TH-001, CL-001 AND BR-001

STATUS: PRELIMINARY ENGINEERING - NOT MACHINING APPROVED

SHEET 5/6 SCALE NTS UNITS mm

REV A INSPECTION NEAR 20 C

Feature	Nominal / tolerance	Inspection
AS568-166 candidate	171.12 ID x 2.62 CS	Supplier cert + optical / calibrated fixture
Main gland	OD 176.35 +0.05/-0; W 3.70 +/-0.05; D 2.12 +/-0.02	CMM/vision OD-position; depth mic; profilometer
AS568-013 candidate	10.82 ID x 1.78 CS	Supplier cert; lot verification
Transfer gland	OD 14.38 +0.05/-0; W 2.50 +/-0.05; D 1.35 +/-0.02	Optical comparator/CMM; depth mic
Rib support plane	3.50 +/-0.03 above insert underside	CMM height map after conditioning
Gasket strip	2.00 +/-0.05 W x 2.00 +/-0.05 free	Incoming thickness/width and force-deflection lot test
Installed gap	1.35 to 1.75	Feeler/replica or displacement mapping under controlled preload
Cover opening	additional <=0.10; local seal <=0.03	Loaded DIC / displacement probes at 30 kPa case

NOTES

1. Use the supplied DXF outlines as preliminary cutting geometry only; preserve separate strips where flow passages must remain open.
2. Nominal 25 percent rib-gasket compression is geometric. Actual compound force, bulging, compression set and durability remain release checks.
3. Flatness, thickness, position and assembled opening are distinct inspections; do not substitute one for another.
4. Drawing dimensions override scale. Ordinary unspecified machined dimensions +/-0.10.

SEALS, GASKETS AND INSPECTION PLAN

STATUS: PRELIMINARY ENGINEERING - NOT MACHINING APPROVED

SHEET 6/6 SCALE NTS UNITS mm

REV A INSPECTION NEAR 20 C